University of Palermo

Computer Engineering

Analysis

Math Formulary

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Introduction

I'm proud to introduce you to an amazing formulary that contains almost all the argument that you can face up at university.

This is not the best book you can buy but the only one that can allow you to pass an university (or college) exam.

Note: To indicate a multiplication, any sign has often been omitted, but sometimes it is possible to contract a point in cases where it was necessary to make the operation more legible (and therefore less misunderstood).

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1 Algebra

1.1 Exp. Rules

Exp. Rules		
Operation	Result	Note
$a^m \cdot a^n$	a^{m+n}	$NONE$
$a^m : a^n$	a^{m-n}	$a \neq 0$
$(a^m)^n$	$a^{m \cdot n}$	$NONE$
a^{-n}	$\frac{1}{a^n}$	$a \neq 0$

1.2 Products

Products		
Operation	Result	Note
$(a \pm b)^2$	$a^2 \pm 2ab + b^2$	$NONE$
$(a + b + c)^2$	$a^2 + b^2 + c^2 + 2ab + 2ac + 2bc$	$NONE$
$(a \pm b)^3$	$a^3 \pm 3a^2b + 3ab^2 \pm b^3$	$NONE$

1.2.1 Pascal's Triangle

$n = 0$						1						(a+b)^0
$n = 1$					1		1					(a+b)^1
$n = 2$				1		2		1				(a+b)$^2 = 1 \cdot a^2 + 2 \cdot ab + 1 \cdot b^2$
$n = 3$			1		3		3		1			(a+b)^3
$n = 4$		1		4		6		4		1		(a+b)^4
$n = n + 1$	1		1+4		4+6		6+4		5		4+1	1

1.3 Scomposition

Scomposition		
Operation	Result	Note
$a^2 - b^2$	$(a+b)(a-b)$	$NONE$
$a^+ \pm b^3$	$(a \pm b)(a^2 \mp ab + b^2)$	$NONE$
$x^2 + (a+b)x + ab$	$(x+a)(x+b)$	$NONE$

1.4 Radicals

Radicals		
Operation	Result	Note
$\sqrt[n]{ab}$	$\sqrt[n]{a} \cdot \sqrt[n]{b}$	$NONE$
$\sqrt[n]{a:b}$	$\sqrt[n]{a} : \sqrt[n]{b}$	$NONE$
$a^{\frac{m}{n}}$	$\sqrt[n]{a^m}$	$\equiv\left(\sqrt[n]{a}\right)^m$
$\frac{1}{\sqrt{a}}$	$\frac{\sqrt{a}}{a}$	$NONE$
$\sqrt[n]{a^n}$	a	$if\ n\ odd$
$\sqrt[n]{a^n}$	$\lvert a \rvert$	$if\ n\ even$
$\sqrt{a \pm \sqrt{b}}$	$\sqrt{\frac{a+\sqrt{a^2-b}}{2}} \pm \sqrt{\frac{a-\sqrt{a^2-b}}{2}}$	$NONE$

1.5 Logarithm

Logarithm		
Operation	Result	Note
$\log_a 1$	0	$a > 0, a \neq 1$
$\log_a a$	1	$NONE$
$a^{\log_a b}$	a	$NONE$
$\log_a(b \cdot c)$	$\log_a b + \log_a c$	$b > 0, c > 0$
$\log_a \frac{b}{c}$	$\log_a b - \log_a c$	$b > 0, c > 0$
$\log_a b^n$	$n \cdot \log_a b$	$b > 0, n \in \mathbb{R}$
$\log_a b$	$\frac{\log_c b}{\log_c a}$	$a \neq 1, c \neq 1, a > 0, b > 0, c > 0$

2 Trigonometry

2.1 Common degree value

Exp. Rules					
Degree	Rad	sin	cos	tg	cotg
$0°$	0	0	1	0	∞
$15°$	$\frac{\pi}{12}$	$\frac{\sqrt{6}-\sqrt{2}}{4}$	$\frac{\sqrt{6}+\sqrt{2}}{4}$	$2-\sqrt{3}$	$2+\sqrt{3}$
$30°$	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$	$\sqrt{3}$
$45°$	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1	1
$60°$	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$	$\frac{\sqrt{3}}{3}$
$75°$	$\frac{5\pi}{12}$	$\frac{\sqrt{6}+\sqrt{2}}{4}$	$\frac{\sqrt{6}-\sqrt{2}}{4}$	$2+\sqrt{3}$	$2-\sqrt{3}$
$90°$	$\frac{\pi}{2}$	1	0	∞	0
$180°$	π	0	-1	0	∞
$270°$	$\frac{3}{2}\pi$	-1	0	∞	0
$360°$	2π	0	1	0	∞

2.1.1 Addition and subtraction

Addition and subtraction		
Function	Addition	Subtraction
sin	$sin(\alpha+\beta) = sin\alpha cos\beta + cos\alpha sin\beta$	$sin(\alpha-\beta) = sin\alpha cos\beta - cos\alpha sin\beta$
cos	$cos(\alpha+\beta) = cos\alpha cos\beta - sen\alpha sin\beta$	$cos(\alpha-\beta) = cos\alpha cos\beta + sen\alpha sin\beta$
tg	$tg(\alpha+\beta) = \dfrac{tg\alpha + tg\beta}{1 - tg\alpha \cdot tg\beta}$	$tg(\alpha-\beta) = \dfrac{tg\alpha - tg\beta}{1 + tg\alpha \cdot tg\beta}$
$Note:$ $tg(\alpha \pm \beta)$	$\alpha+\beta \neq \dfrac{\pi}{2} + k\pi, \alpha \neq \dfrac{\pi}{2} + k\pi, \beta \neq \dfrac{\pi}{2} + k\pi$	$\alpha-\beta \neq \dfrac{\pi}{2} + k\pi, \alpha \neq \dfrac{\pi}{2} + k\pi, \beta \neq \dfrac{\pi}{2} + k\pi$

2.1.2 Duplication and bisection

Duplication and bisection			
Function	Duplication	Bisection	Note
sin	$sen(2\alpha) = 2sen\alpha cos\alpha$	$sin\frac{\alpha}{2} = \pm\sqrt{\frac{1-cos\alpha}{2}}$	$NONE$
cos	$cos(2\alpha) = cos^2\alpha - sen^2\alpha$	$cos\frac{\alpha}{2} = \pm\sqrt{\frac{1+cos\alpha}{2}}$	$NONE$
tg	$tg(2\alpha) = \frac{2tg\alpha}{1-tg^2\alpha}$	$tg\frac{\alpha}{2} = \pm\sqrt{\frac{1-cos\alpha}{1+cos\alpha}}$	$\alpha \neq \pi + 2k\pi$ $\pm\, depends\, of\, the\, position\, of\, \alpha$ $in\, the\, dial$

2.1.3 Prosthaphaeresis

Prosthaphaeresis	
Operation	Result
$sin\, p + sin\, q$	$2sin\frac{p+q}{2} \cdot cos\frac{p-q}{w}$
$sin\ - sin\, q$	$2cos\frac{p+q}{2} \cdot sen\frac{p-q}{w}$
$cos\, p + sin\, q$	$2cos\frac{p+q}{2} \cdot cos\frac{p-q}{w}$
$cos\, p + sin\, q$	$-2sin\frac{p+q}{2} \cdot sen\frac{p-q}{w}$

2.1.4 Werner

Werner	
Operation	Result
$sin\alpha\ sin\beta$	$\frac{1}{2}\Big[cos(\alpha-\beta)-cos(\alpha+\beta)\Big]$
$cos\alpha\ cos\beta$	$\frac{1}{2}\Big[cos(\alpha+\beta)+cos(\alpha-\beta)\Big]$
$sin\alpha\ cos\beta$	$\frac{1}{2}\Big[sin(\alpha+\beta)+sin(\alpha-\beta)\Big]$

3 Geometry

3.1 Distance Between Points

Distance B. Points		
Operation	Result	Note
$\overline{AB}$	$\sqrt{(x_b - x_a)^2 + (y_b - y_a)^2}$	$A(x_a, y_a), B(x_b, y_b)$
$\overline{AB}$	$\lvert y_b - y_a \rvert$	$y \parallel x$
$\overline{AB}$	$\lvert x_b - x_a \rvert$	$x \parallel y$
$\overline{OP}$	$\sqrt{x_p^2 + y_p^2}$	$P(0, 0)$
$\overline{M}(x_m, y_m)$	$x_m = \dfrac{x_a + x_b}{2}, y_m = \dfrac{y_a + y_b}{2}$	$NONE$

4 Calculus

4.1 Limits

4.1.1 Infinity Order

Infinity Order	
Notation	Note
n^n	$Fastest$
$n!$	
an	$a > 1$
n^α	$\alpha > 0$
$log\, n$	$Slowest$

4.1.2 Infinitesimal

Limits (infintesimal)		
Limits	Results	Note
$\lim_{x\to 0} sin\, x$	$\equiv x$	
$\lim_{x\to 0} tg\, x$	$\equiv x$	
$\lim_{x\to 0} arctan\, x$	$\equiv x$	
$\lim_{x\to 0} arcsin\, x$	$\equiv x$	
$\lim_{x\to 0} e^x - 1$	$\equiv x$	
$\lim_{x\to 0} ln(1+x)$	$\equiv x$	
$\lim_{x\to 0} ln(1-x)$	$\equiv -x$	
$\lim_{x\to 0} \sqrt[n]{1+x} - 1$	$\equiv \frac{x}{n}$	
$\lim_{x\to 0} (1+x)^k - 1$	$\equiv kx$	
$\lim_{x\to 0} a^x - 1$	$\equiv x\, ln\, a$	
$\lim_{x\to 0} 1 - cos\, x$	$\equiv \frac{1}{2}x^2$	

4.1.3 Taylor-Mc Laurin Approximation

Taylor-Mc Laurin Approximation		
Function	Approximation	Center
e^x	$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots + \frac{x^n}{n!} + o(x^n)$	$x \to 0$
$ln(1+x)$	$x - \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + (-1)^{n-1}\frac{x^n}{n} + o(x^n)$	$x \to 0$
$(1+x)^\alpha$	$1 + \alpha x + \binom{\alpha}{2}x^2 + \binom{\alpha}{3}x^3 + \cdots + \binom{\alpha}{n}x^n + o(x^n)$	$x \to 0$
$\frac{1}{1+x}$	$1 - x + x^2 - x^3 + \cdots + (-1)^n x^n + o(x^n)$	$x \to 0$
$\frac{1}{1-x}$	$1 + x + x^2 + x^3 + \cdots + x^n + o(x^n)$	$x \to 0$
$\sqrt{1+x}$	$1 + \frac{1}{2}x + \frac{1}{8}x^2 + \frac{1}{16}x^3 + \cdots + \binom{\frac{1}{2}}{n}x^n + o(x^n)$	$x \to 0$
$\frac{1}{\sqrt{1+x}}$	$1 - \frac{1}{2}x + \frac{3}{8}x^2 - \frac{5}{16}x^3 + \cdots + \binom{-\frac{1}{2}}{n}x^n + o(x^n)$	$x \to 0$
$\sqrt[3]{1+x}$	$1 + \frac{1}{3}x - \frac{1}{9}x^2 5\frac{5}{81}x^3 + \cdots + \binom{\frac{1}{3}}{n}x^n + o(x^n)$	$x \to 0$
$\frac{1}{\sqrt[3]{1+x}}$	$1 - \frac{1}{3}x + \frac{2}{9}x^2 - \frac{7}{81}x^3 + \cdots + \binom{-\frac{1}{3}}{n}x^n + o(x^n)$	$x \to 0$

Taylor-Mc Laurin Approximation (Trigonometric)		
Function	Approximation	Center
$sin\, x$	$x - \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots + \frac{x^{2n+1}}{(2n+1)!} + o(x^{2n+2})$	$x \to 0$
$cos\, x$	$1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots + \frac{x^2 n}{2n!} + o(x^{2n+1})$	$x \to 0$
$tg\, x$	$x + \frac{x^3}{3!} + \frac{2}{15}x^5 + o(x^6)$	$x \to 0$
$arcsin\, x$	$x + \frac{x^3}{6!} + \frac{3}{40}x^5 + o(x^6)$	$x \to 0$
$arctg\, x$	$x - \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots + \frac{x^{2n+1}}{2n+1} + o(x^{2n+2})$	$x \to 0$
$sinh\, x$	$x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots + \frac{x^{2n+1}}{(2n+1)!} + o(x^{2n+2})$	$x \to 0$
$cosh\, x$	$1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots + \frac{x^{2n}}{2n!} + o(x^{2n+1}$	$x \to 0$
$tgh\, x$	$x - \frac{x^3}{3!} + \frac{2}{15}x^5 + o(x^6)$	$x \to 0$
$sett\, tgh\, x$	$x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots + \frac{x^{2n+1}}{2n+1} + o(x^{2n+2})$	$x \to 0$

4.1.4 Goniometric

Limits (Goniometric)		
Limits	Results	Note
$\lim_{x \to 0} \frac{sin\, x}{x}$	1	
$\lim_{x \to 0} \frac{1 - cos\, x}{x}$	0	
$\lim_{x \to 0} \frac{tg\, x}{x}$	1	
$\lim_{x \to 0} \frac{arcsin\, x}{x}$	1	
$\lim_{x \to 0} \frac{arctg\, x}{x}$	1	
$\lim_{x \to 1} \frac{arccos\, x}{\sqrt{1 - x}}$	$\sqrt{2}$	

4.1.5 Exp. and Log

Limits (Exp. and Log)		
Limits	Results	Note
$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{x}$	e	$Increase$
$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{x+1}$	e	$Decrease$
$\lim_{x\to\infty}\left(1-\frac{1}{x}\right)^{x}$	$\frac{1}{e}$	$Increase$
$\lim_{x\to\infty}\left(1+\frac{k}{x}\right)^{x}$	e^k	$Increase$
$\lim_{x\to 0}(1+x)^{\left(\frac{1}{x}\right)}$	e	
$\lim_{x\to 0}(1-x)^{\left(\frac{1}{x}\right)}$	$\frac{1}{e}$	
$\lim_{x\to 0}\frac{a^x-1}{x}$	$ln\, a$	
$\lim_{x\to 0}\frac{e^x-1}{x}$	1	
$\lim_{x\to 0}\frac{(1+x)^a-1}{x}$	a	
$\lim_{x\to 0}\frac{\log_a(1+x)}{x}$	$\log_a e$	
$\lim_{x\to 0}\frac{\ln(1\pm x)}{x}$	± 1	
$\lim_{x\to 0^+} x^a ln\, x$	0	$a>0$
$\lim_{x\to +\infty}\frac{ln\, x}{a^x}$	0	$a>0$
$\lim_{x\to +\infty}\frac{ln\, x}{x^a}$	0	$a>0$
$\lim_{x\to +\infty}\frac{x^a}{a^x}$	0	$a>0$
$\lim_{x\to 0}\frac{b^{(nx)}-1}{x}$	$n\, ln\, b$	$a>0$

$\lim_{x\to -\infty} x^a e^x$	0	$\forall a\in\mathbb{R}^+$

De L'Hopital Trick

$$if\, are\, verified \Rightarrow \begin{cases} f(x), g(x)\, continuous \\ f(x), g(x)\, derivable \\ \exists \lim_{x \to x_0} \frac{f'(x)}{g'(x)} \end{cases}$$

Indetermined Form $[0 \cdot \infty] \Rightarrow \lim_{x \to x_0} f(x) \cdot g(x) \to \lim_{x \to x_0} \frac{f(x)}{\frac{1}{g(x)}} = \frac{\infty}{\infty}$

Indetermined Form $[0^0] \Rightarrow \lim_{x \to x_0} f(x)^{g(x)} = \lim_{x \to x_0} e^{g(x)\, ln \left[f(x) \right]}$

4.2 Derivative

4.2.1 Elementary

DERIVATIVE (Elementary functions)		
Primitive	Derivate	Note
k	0	$NONE$
x	1	$NONE$
$\|x\|$	$\frac{\|x\|}{x}$	$NONE$
x^a	ax^{a-1}	$a \in \mathbb{R} - [0,1]$
$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$	$x > 0$
$\sqrt[n]{x^m}$	$\frac{m}{n\sqrt[n]{x^{n-m}}}$	$x > 0, n \in \mathbb{N}, n > m$
a^x	$a^x \, lna$	$x > 0$
$\frac{1}{x}$	$-\frac{1}{x^2}$	$NONE$
$\frac{1}{x^\alpha}$	$-\frac{\alpha}{x^{(\alpha+1)}}$	$NONE$
e^x	e^x	$NONE$
$e^{k\,x}$	$k\,e^{k\,x}$	$NONE$
$ln\,x$	$\frac{1}{x}$	$Natural\ logaritm$
$log^\alpha x$	$\alpha\frac{log^{(\alpha-1)}x}{x}$	$NONE$
$\log_a x$	$\frac{1}{x}\frac{1}{ln\,a}$	$NONE$

4.2.2 Trigonometric

DERIVATIVE (Trigonometric)		
Primitive	Derivate	Note
$sin\,x$	$cos\,x$	
$sin\,x^\circ$	$cos\,x\frac{\pi}{180^\circ}$	
$cos\,x$	$-sin\,x$	
$cotg\,x^\circ$	$-sin\,x\frac{\pi}{180^\circ}$	
$tg\,x$	$\frac{1}{cos^2x}$	
$Alternateform$	$1+tg^2x$	
$cotg\,x$	$-\frac{1}{sin^2x}$	
$Alternateform$	$-(1+cotg^2x)$	
$arcsin\,x$	$\frac{1}{\sqrt{1-x^2}}$	
$arccos\,x$	$-\frac{1}{\sqrt{1-x^2}}$	
$arctg\,x$	$\frac{1}{1+x^2}$	
$arccotg\,x$	$-\frac{1}{x^2+1}$	
$sec\,x$	$tg\,x\ sec\,x$	
$cosec\,x$	$-cotg\,x\ cosec\,x$	
$sinh\,x$	$cosh\,x$	
$cosh\,x$	$sinh\,x$	

DERIVATIVE (Trigonometric)		
$tgh\,x$	$\frac{1}{cosh^2x}$	
$cotgh\,x$	$\frac{1}{sinh^2x}$	
$settsinh\,x$	$\frac{1}{\sqrt{x^2+1}}$	
$settcosh\,x$	$\frac{1}{\sqrt{x^2-1}}$	
$setttgh\,x$	$\frac{1}{1-x^2}$	
$settcotgh\,x$	$-\frac{1}{x^2-1}$	

4.2.3 Composite Functions

DERIVATIVE (Composite Functions)		
Primitive	Derivate	Note
$k\,f(x)$	$k\,f'(x)$	$Derivative \equiv f'(x)$
$\lvert f(x)\rvert$	$\frac{\lvert fx\rvert}{fx} f'(x)$	
$\sqrt{f(x)}$	$\frac{f'(x)}{2\sqrt{f(x)}}$	
$\sqrt[n]{f(x)^m}$	$\frac{m}{n\sqrt[n]{f(x)^{n-m}}}$	
$\frac{f(x)}{k}$	$\frac{f'(x)}{k}$	$k \neq 0$
$\frac{k}{f(x)}$	$-\frac{kf'(x)}{[f(x)]^2}$	
$e^{f(x)}$	$f'(x)e^{f(x)}$	
$a^{f(x)}$	$f'(x)a^{f(x)} ln\, a$	
$f(x)^{\alpha}$	$\alpha f'(x)[f(x)]^{(\alpha-1)}$	
$f(x) \pm g(x)$	$f'(x) \pm g'(x)$	
$f(x)\,g(x)$	$f'(x)\,g(x) + f(x)\,g'(x)$	
$\frac{f(x)}{g(x)}$	$\frac{f'(x)\,g(x) - f(x)\,g'(x)}{[g(x)]^2}$	
$f(x)^{g(x)}$	$f(x)^{g(x)} \cdot [g'(x)\,ln(f(x)) + \frac{f'(x)}{f(x)} g(x)]$	
$f(g(h(x)))$	$f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x)$	
$sin(f(x))$	$f'(x)sin(fx)$	$All\,Trigon.\,equals$
$tg(f(x))$	$\frac{f'(x)}{1+f^2(x)}$	$All\,Trigon.\,equals$

4.3 Function Study

4.3.1 Study Process

1. $Domain$
2. $Simmetry$
3. $Positivity$
4. $Intersection$
5. $Limits\, and\, asymptote$
6. $Continuity$
7. $I\, Derivative\, (f')$
8. $Stationary\, points\, +\, min/max$
9. $Lipschitzianità$
10. $II\, Derivative\, (f'')$
11. $Uniform\, continuity$
12. $Graphic$
13. $Definite\, integral$

Domain

1. $denominator \neq 0$
2. $\sqrt[n]{x} \rightarrow if\, n = pair \Rightarrow x \geq 0$
3. $log\, x \Rightarrow x > 0$
4. $f(x)^{g(x)} \Rightarrow f(x) > 0$
5. $arcsin\, x\ or\ arccos\, x \Rightarrow -1 \leq x \leq 1$

Simmetry

1. $f(-x) = f(x) \Rightarrow f(x)\, is\, pair$
2. $f(-x) = -f(x) \Rightarrow f(x)\, is\, odd$
3. $f(-x) = f(x) \Rightarrow f(x)\, isn't\, pair\, or\, odd$

Positivity

1. $f(x) \geq 0$

Intersection

1. $f(0) = y \Rightarrow A(0, y)$
2. $f(x) = 0 \Rightarrow B(x, 0)$

   Note: It is useful to draw the graph now to get a clear idea of the function

Limits and asymptote

Limits should be calculated at the ends of the domain and at the accumulation points

1. $if\ \exists \lim_{x \to \pm\infty} f(x) = c \in \mathbb{R} \Rightarrow y = c\ \ is\, a\, horizontal\, asymptote$
2. $if\ \exists \lim_{x \to x_0^{\pm}} f(x) = \pm\infty\ \ \Rightarrow x = x_0\ t\, is\, a\, vertical\, asymptote$
3. $if\ \exists \lim_{x \to \pm\infty} f(x) = c$

   $if\, c \in \mathbb{R} \Rightarrow\ no\, oblique\, asymptote$

   $if\, c = \pm\infty \Rightarrow\ if\ \exists \lim_{x \to \pm\infty} \dfrac{f(x)}{x} = m\ \in\ \mathbb{R} - \{0\}$

   $if\ \exists \lim_{x \to \pm\infty} f(x) - mx = q\ \in\ \mathbb{R} \Rightarrow y = mx + q\, is\, an\, oblique\, asymptote$

   Note: the oblique asymptote can exists if and only if there is no horizontal asymptote

Continuity

1. $if\ \exists \lim = l_1 \in \mathbb{R}\ and\ \exists \lim = l_2 \in \mathbb{R}\ but\ l_1 \neq l_2 \Rightarrow\ \ I\ species\ discontinuity$
2. $if\ \nexists \lim\ \ or\ \exists \lim = \pm\infty \Rightarrow\ \ II\ species\ discontinuity$
3. $if\ \exists \lim_{x \to x_0} = l_1 \in \mathbb{R}\ and\ \exists \lim = l_2 \in \mathbb{R}\ with\ l_1 = l_2; but\ f(x_0) \neq l_1\ (or\ l_2) \Rightarrow$

   $species\ discontinuity\ (eliminable)$

I Derivative

$$f'(x) \Rightarrow \begin{cases} f'(x) > 0, & f(x)\,\text{it is growing} \\ f'(x) < 0, & f(x)\,\text{it is decreasing} \end{cases}$$

Relative Min and Max

$$x_0\, relative\, min \Leftrightarrow f'(x) = 0\, and \begin{cases} f'(x) < 0, & for\, x \in\,]a, x_0[\\ f'(x) = 0, & \\ f'(x) > 0, & for\, x \in]x_0, b[\end{cases}$$

$$x_0\, relative\, max \Leftrightarrow f'(x) = 0\, and \begin{cases} f'(x) > 0, & for\, x \in\,]a, x_0[\\ f'(x) = 0, & \\ f'(x) < 0, & for\, x \in]x_0, b[\end{cases}$$

Successive derivative method

$if\, \exists f^n(x) = 0 \Rightarrow calculate f^{n+1}(x)$

$$if (n+1)\, is\, pair \begin{cases} f^{n+1}(x) < 0, & \exists\, Relative\, Minimum \\ f^{n+1}(x) > 0, & \exists\, Relative\, Maximum \end{cases}$$

$$if (n+1)\, is\, odd \begin{cases} f^{n+1}(x) > 0, & \exists \text{ an ascending inflection with a horizontal tangent} \\ f^{n+1}(x) < 0, & \exists \text{ a discending inflection with a horizontal tangent} \end{cases}$$

Points of non-derivability

1. $if\ \exists \lim\limits_{x_0^{\pm}} = l_1, l_2 \in \mathbb{R}\ but\ l_1 \neq l_2 \Rightarrow\ x_0\, is\, an\, angular\, point$
2. $if\ \exists \lim\limits_{x_0^{\pm}} = l_1, l_2\, (\pm\infty)\ but\ l_1 \neq l_2 \Rightarrow\ x_0\, is\, a\, cusp$
3. $if\ \exists \lim\limits_{x_0^{\pm}} = l_1, l_2\, (\pm\infty)\ and\ l_1 = l_2 \Rightarrow\ x_0$ is a flexion with a vertical tangent

Absolute Max and Min

$\exists\, Max\, and\, Min\ \Leftrightarrow f(x)\, is\, continue\, in\, a\, domain\, closed\, and\, limitated\ \rightarrow f(x)\, continue\, in\, [a, b]$

$$if \begin{cases} \lim_{x\to a,b} f(x) = \pm\infty\ \Rightarrow NO\ MIN\ OR\ MAX, , \\ \lim_{x\to a,b} f(x) = l_1, l_2 \in \mathbb{R} \begin{cases} f(x)' \neq 0 \forall x \in dom\, f(x)\ \Rightarrow NO\ MIN\ OR\ MAX, \\ \exists f'(x) = 0\ (Take\, this\, point\, x_0) \begin{cases} f''(x_0) = n > 0, & \text{ABS. MINIMUM} \\ f''(x_0) = n < 0, & \text{ABS. MAXIMUM} \end{cases} \end{cases} \end{cases}$$

II Derivative - Concavity

1. $f''(x) = 0 \Rightarrow$ candidates for inflection points
2. $f''(x) =< 0 \Rightarrow f(x)$ concave $[\cap]$
3. $f''(x) > 0 \Rightarrow f(x)$ convex $[\cup]$

Concavity by limits

Concavity by limits (m=$0^{\pm} or\ m = \pm\infty$)			
$x \to$	$lim f(x)$	$lim \frac{f(x)}{x}$	Concavity
$+\infty$	$+\infty$	0^+	$\cap$
$+\infty$	$+\infty$	$+\infty$	$\cup$
$-\infty$	$+\infty$	0^-	$\cap$
$-\infty$	$+\infty$	$-\infty$	$\cup$
$-\infty$	$-\infty$	0^+	$\cup$
$-\infty$	$-\infty$	$+\infty$	$\cap$
$+\infty$	$-\infty$	0^-	$\cup$
$+\infty$	$-\infty$	$-\infty$	$\cap$

Concavity by limits (m$\neq$ 0)			
$x \to$	$lim f(x)$	q	Concavity
$+\infty$	$+\infty$	$+\infty$	$\cap$
$+\infty$	$-\infty$	$-\infty$	$\cup$
$-\infty$	$+\infty$	$+\infty$	$\cap$
$-\infty$	$-\infty$	$-\infty$	$\cup$

1. $lim_{x \to x_0^+} = +\infty \Rightarrow \cup$
2. $lim_{x \to x_0^-} = -\infty \Rightarrow \cap$
3. $lim_{x \to x_0^+} = -\infty \Rightarrow \cap$
4. $lim_{x \to x_0^-} = +\infty \Rightarrow \cup$

Note: to use this rules it's not necessary to calculate the second derivative but it has to exists!!! Generally it is not recommended to use this method because it may not accurately represent the whole curve trend but mater artium necessitas (Necessity is the mother of invention).

Continuity

Supposed

$$f : A(\subseteq \mathbb{R}) \to \mathbb{R}$$

1. $$f\, is\, continuous\, in\, x \in A\, if\ \ \forall \epsilon > 0,\, \exists \delta > 0\ \mid\ \forall y \in A\, |x-y| < \delta \Rightarrow |f(x)-f(y)| < \epsilon$$
2. $$f\, is\, continuous\, in\, A\, if\ \ \forall x \in A, \epsilon > 0,\, \exists \delta > 0\ \mid\ \forall y \in A\, |x-y| < \delta \Rightarrow |f(x)-f(y)| < \epsilon$$
3. $$f\, is\, uniformly\, continuous\, in\, A\, if\ \ \forall \epsilon > 0,\, \exists \delta > 0\ \mid\ \forall x, y \in A\, |x-y| < \delta \Rightarrow |f(x)-f(y)| < \epsilon$$
4. $$f\, is\, Lipschitziana\, if\ \ \exists\, L > 0\ \mid\ \forall x, y \in A\, |f(x) - f(y)| \leq L|x-y|$$

4.3.2 From Derivative to Primitive

Foglio1

Derivative [f'(x) f(t)]	Primitive [(F(x))]
In the intervals where f(t) >0	F(x) it is growing
In the intervals where f(t) <0	F(x) it is decreasing
In the points where f(t)=0 and change sign in the neighborhood	F(x) has, in the same points, maximum or minimum
In the points where f(t)=0 and does not change sign in the neighborhood	F(x) has a horizontal tangent flexion
If f(t) is growing	F(x) has concavity upward
If f(t) is decreasing	F(x) has concavity down
In the points where f(t) has horizontal tangents	F(x) has, in the same points, inflections
If f(t) tends to more or less infinite on the right and on the left side of a point	F(x) will have, at that point, a bend with a vertical tangent
In points where f(t) has a maximum	F(x) has a descending flexion
In points where f(t) has a minimum	F(x) has an ascending flexion
In the points where f(t) has an ascending flexion	F(x) immediately after grows faster
In the points where f(t) has a descending flexion	F(x) soon after it grows more slowly
In the points where f(t) is not derivable	F(x) has a cusp or a corner point or a vertical asymptote or a vertical tangent bend
If f(t) is even	F(x) is odd
If f(t) is odd	F(x) is even
In the points where f(t) is not derivable in a point	F(x) can still be continued at that point

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4.3.3 Natural Domain

Function	Domain
$\frac{k}{x}$	$x \neq 0$
$\sqrt[n]{x}$	$x \geq 0\,(n\,pair)$
$\sqrt[n]{x}$	$\mathbb{R}\,(n\,odd)$
$log_a b$	$a > 0, a \neq 1, b > 0$
$a^{f(x)}$	$a > 0, a \neq 1$
$sin\,x$	$\mathbb{R}$
$cos\,x$	$\mathbb{R}$
$tg\,x$	$\mathbb{R} - \{\frac{\pi}{2} + k\pi\}$
$arcsin\,x$	[-1,1]
$arccos\,x$	[-1,1]
$arctg\,x$	$\mathbb{R}$
$arccotg\,x$	$\mathbb{R}$

4.4 Integrals

4.5 Property

1. Properties of the integration extremes
   $$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$$
2. Properties of the overlapping ends
   $$\int_a^a f(x)\,dx = 0$$
3. Linearity of the integrals with respect to the extremes
   $$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx\, with\, c \in [a,b]$$
4. Additivity of the integral
   $$\int_a^b \big[f(x) + g(x)\big]dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$$
5. Properties of the integration extremes
   $$\int_a^b \big[kf(x)\big]\,dx = k\int_a^b f(x)\,dx$$

4.6 Torricelli - Barrow

$$\int_a^b f(x)\,dx = \Big[F(x)\Big]_a^b = F(b) - F(a)$$

4.6.1 Elementary

Integrals (Elementary)		
Integrals	Primitive	Note
$\int k\,dx$	$kx + c$	
$\int x^n\,dx$	$\frac{1}{n+1} \cdot x^{n+1}$	
$\int \sqrt{x}\,dx$	$\frac{2}{3}x\sqrt{x}$	$n \neq -1$
$\int a^x\,dx$	$a^x \cdot \frac{1}{ln\,a} + c$	
$\int e^x\,dx$	$e^x + c$	
$\int \frac{1}{x}\,dx$	$ln\lvert x\rvert + c$	
$\int \frac{a}{ax+b}\,dx$	$\frac{1}{b}ln\lvert ax+b\rvert\,dx$	
$\int \frac{1}{a^2-x^2}\,dx$	$\frac{1}{2a}ln\left\lvert\frac{a+x}{a-x}\right\rvert + c$	
$\int \frac{1}{a^2+x^2}\,dx$	$\frac{1}{a}arctg\left(\frac{x}{a}\right) + c$	
$\int \frac{1}{a^2-x^2}\,dx$	$\frac{1}{2a}ln\left\lvert\frac{x-a}{x+a}\right\rvert + c$	
$\int \frac{1}{\sqrt{x^2 \pm a^2}}\,dx$	$ln\left\lvert x + \sqrt{x^2 \pm a^2}\right\rvert + c$	
$\int \frac{1}{1-x^2}\,dx$	$arcsin\,x + c$	
$\int \sqrt{a^2-x^2}\,dx$	$\frac{1}{2}a^2arcsin\left(\frac{x}{a}\right) + \frac{1}{2}x\sqrt{a^2-x^2} + c$	

4.6.2 Goniometric

Integrals (Goniometric)		
Integrals	Primitive	Note
$\int sin\,x\,dx$	$-cos\,x + c$	
$\int cos\,x\,dx$	$sin\,x + c$	
$\int tg\,xdx$	$-ln\lvert cos\,x\rvert + c$	
$\int cotg\,x\,dx$	$ln\lvert sinx\rvert + c$	
$\int arctg\,x\,dx$	$x\,arctg(x) - \frac{1}{2}ln(1+x^2) + c$	
$\int \frac{1}{cos^2x}\,dx$	$tg\,x + c$	
$\int \frac{1}{sin^2x}\,dx$	$-cotg\,x + c$	
$\int \frac{1}{c\,osx}\,dx$	$ln\left\lvert cotg\left(\frac{\pi}{4} - \frac{x}{2}\right)\right\rvert + c$	
$\int \frac{1}{sin\,x}\,dx$	$ln\left\lvert tg\left(\frac{x}{2}\right)\right\rvert + c$	
$\int \frac{1}{1+cos\,x}\,dx$	$ln\left\lvert tg\left(\frac{x}{2}\right)\right\rvert + c$	

Integrals (Goniometric)		
Integrals	Primitive	Note
$\int sinh\, x\, dx$	$cosh\, x + c$	
$\int cosh\, x\, dx$	$sinh\, x$	
$\int tgh\, x\, dx$	$ln\lvert cosh\, x\rvert + c$	
$\int cotgh\, x\, dx$	$ln\lvert sinh\, x\rvert + c$	
$\int sin^2\, x\, dx$	$\frac{1}{2}(x - sin\, x\, cos\, x) + c$	
$\int sin^2\, x\, dx$	$\frac{1}{2}(x + sin\, x\, cos\, x) + c$	
$\int tg^2\, x\, dx$	$tg\, x - x + c$	
$\int cotg^2\, x\, dx$	$-x - cotg\, x + c$	
$\int sin(\alpha x)\, dx$	$-\frac{1}{\alpha} cos(\alpha x) + c$	
$\int cos(\alpha x)\, dx$	$\frac{1}{\alpha} sin(\alpha x) + c$	
$\int \frac{1}{sin^2 x\, cos^2 x}\, dx$	$tg\, x - cotg\, x + c$	

4.6.3 Composite Functions

Integrals (Composite Functions)		
Integrals	Primitive	Note
$\int f(x)^n \cdot f'(x)\,dx$	$\frac{1}{n+1} f(x)^{n+1} + c$	$f'(x) \equiv Derivative[f(x)]$
$\int a^{f(x)} \cdot f'(x)$	$\frac{1}{n+1} f(x)^{n+1} + c$	
$\int e^{f(x)}\,dx$	$e^{f(x)} + c$	
$\int \frac{1}{f(x)} \cdot f'(x)\,dx$	$ln\lvert f(x)\rvert + c$	
$\int sin\big(f(x)\big) \cdot f'(x)\,dx$	$-cos\big(f(x)\big) + c$	
$\int cos\big(f(x)\big) \cdot f'(x)\,dx$	$sin\big(f(x)\big) + c$	
$\int tg\big(f(x)\big) \cdot f'(x)\,dx$	$-ln\big\lvert cos\big(f(x)\big)\big\rvert + c$	
$\int cotg\big(f(x)\big) \cdot f'(x)\,dx$	$ln\big\lvert sin\big(f(x)\big)\big\rvert + c$	
$\int \frac{1}{cos^2\big(f(x)\big)} \cdot f'(x)\,dx$	$tg\big(f(x)\big) + c$	
$\int \frac{1}{sin^2\big(f(x)\big)} \cdot f'(x)\,dx$	$-cotg\big(f(x)\big) + c$	
$\int \frac{1}{a^2 + f^2(x)} \cdot f'(x)\,dx$	$\frac{1}{a} arctg\left(\frac{f(x)}{a}\right) + c$	
$\int \frac{1}{\sqrt{1 - f^2(x)}} \cdot f'(x)\,dx$	$arcsin\,\big(f(x)\big) + c$	
$\int \frac{1}{1 + f^2(x)} \cdot f'(x)\,dx$	$arctg\,\big(f(x)\big) + c$	

$\int sinh\big(f(x)\big) \cdot f'(x)\, dx$	$cosh\big(f(x)\big) + c$	
$\int cosh\big(f(x)\big) \cdot f'(x)\, dx$	$sinh\big(f(x)\big) + c$	
$\int tgh\big(f(x)\big) \cdot f'(x)\, dx$	$ln\big\lvert cosh\big(f(x)\big)\big\rvert$	
$\int cotgh\big(f(x)\big) \cdot f'(x)\, dx$	$ln\big\lvert sinh\big(f(x)\big)\big\rvert$	

Integrals (Others)		
Integrals	Primitive	Note
$\int cos[\alpha - \beta](\alpha - \beta)\, dx$	$\frac{sin}{2(\alpha+\beta)}\Big[(\alpha+\beta)x\Big] + \frac{sin}{2(\alpha-\beta)}\Big[(\alpha-\beta)x\Big] + c$	
$\int sin(\alpha x)cos(\beta x)\, dx$	$-\frac{cos}{2(\alpha+\beta)}\Big[(\alpha+\beta)x\Big] - \frac{cos}{2(\alpha-\beta)}\Big[(\alpha-\beta)x\Big] + c$	

4.6.4 Parametric

Integrals (Parametric)		
Function	Sostitution	Note
$sin\, x$	$\dfrac{2t}{1+t^2}$	$t = tg\left(\dfrac{x}{2}\right)$
$cos\, x$	$\dfrac{1-t^2}{1+t^2}$	$t = tg\left(\dfrac{x}{2}\right)$
$tg\, x$	$-1+z$	$tg\, x + 1 = z$
$\displaystyle\int \frac{1}{\sqrt{x^2 \pm a^2}}\, dx$ $\displaystyle\int \sqrt{x^2 \pm a^2}\, dx$	$t = x + \sqrt{x^2 \pm \alpha^2}$ $x = \dfrac{t^2 + [-(\alpha^2)]}{2t}$ $\sqrt{x^2 \pm \alpha^2} = \dfrac{t^2 \pm \alpha^2}{2t}$ $dx = \dfrac{t^2 \pm \alpha^2}{2t} dt$	

4.7 Notable improper integrals

1. $\displaystyle\int_a^b \frac{1}{(b-x)^\alpha}\, dx \Rightarrow \begin{cases} diverges^+ \; if \; \alpha \geq 1 \\ converges \; if \; \alpha < 1 \end{cases}$

2. $\displaystyle\int_a^\infty e^{-tx}\, dx \Rightarrow \begin{cases} converges \; if \; t > 0 \\ diverges \; if \; t \leq 0 \end{cases}$

3. $\displaystyle\int_a^\infty \frac{1}{x^\rho}\, dx \Rightarrow \begin{cases} converges \; if \; \rho > 1 \\ diverges \; if \; \rho \leq 1 \end{cases}$

4. $\displaystyle\int \frac{1}{e^x + 1}\, dx \; converges \, because \; < \int \frac{1}{e^x} \, (that \, converges)$

5. $\displaystyle\int \frac{1}{x}\, dx \; diverges \, because \; \geq \int \frac{1}{ln\, x} \, (that \, diverges)$

4.8 Integrals by recurrence

1. Cos

$$\int cos^n x\, dx \Rightarrow C_n = \int cos\, x\, (cos\, x)^{n-1} \rightarrow C_n = \frac{1}{n}(cos\, x)^{n-1} sen\, x + \left(\frac{n-1}{n}\right) C_{n-2}$$

2. Sin

$$\int sen^n x\, dx \Rightarrow S_n = -\frac{1}{n}(sen\, x)^{n-1} cos\, x + \left(\frac{n-1}{n}\right) S_{n-2}$$

3. Tg

$$T_n = \int (tg\, x)^n\, dx \Rightarrow T_n = \frac{(tg\, x)^{n-1}}{n-1} - T_{n-2}$$

4. Cotg

$$C_{Tn} = \int (cotg\, x)^n\, dx \Rightarrow C_{Tn} = -\frac{(cotg\, x)^{n-1}}{n-1} - C_{Tn-2}$$

5. Arcsin

$$\int (arcsin\, x)^n\, dx \Rightarrow B_n = -x\, (arcsin\, x)^n + n\sqrt{1-x^2}(arcsin\, x)^{n-1} - n(n-1)B_{n-2}$$

6. Ln

$$L_n = \int x^\alpha ln^n x\, dx \Rightarrow \left(\frac{x^{\alpha+1}}{\alpha+1}\right) ln^n x - \left(\frac{n}{\alpha+1}\right) L_{n-1}$$

7. $\int x^n sen\, x\, dx \Rightarrow S_n = -x^n\, cos\, x + nx^{n-1} sen\, x - n(n-1)S_{n-2}$

8. $E_n = \int x^n e^{\alpha x}\, dx \Rightarrow E_n = \left(\frac{x^n e^{\alpha x}}{\alpha}\right) - \frac{n}{\alpha} E_{n-1}$

9. $H_n = \int \frac{dx}{(1+x^2)^n} \Rightarrow H_n = \left(\frac{x}{2(n-1)(1+x^2)^{n-1}}\right) + \left(\frac{2n-3}{2n-2}\right) H_{n-1}$

10. $\overline{H_n} = \int \frac{dx}{(x^2-1)^n} \Rightarrow \overline{H_n} = \left(-\frac{x}{2(n-1)(1+x^2)^{n-1}}\right) + \left(-\frac{2n-3}{2n-2}\right) \overline{H_{n-1}}$

4.9 Integral by part

$$\int f(x)\, g'(x)\, dx = f(x) \cdot g(x) - \int f'(x)\, g(x)\, dx$$

TIPS:

1. $\int x^n sen\, x\, dx\,, \int x^n cos\, x\, dx \int x^n e^x\, dx \Rightarrow x^n = f(x)$
2. $\int x^n ln\, x\, dx\,, \int x^n arctg\, x\, dx \int x^n arcsin\, x\, dx \Rightarrow x^n = g(x)$
3. $\int ln\, x\, dx\,, \int arctg\, x\, dx \int arcsin\, x\, dx \Rightarrow x^0 \equiv (1 \cdot dx) = g(x)$

4.10 Integral by sostitution

$$\int f(x)\, dx = \int f\big(g(x)\big) \cdot g'(x)\, dx$$

4.11 Integrals of fractional rational functions

1. $$deg(N(x)) > deg(D(x))$$

$$\int \frac{Nx}{Dx}\, dx$$

$$\frac{N(x)}{D(x)} = Q(x) + \frac{R(x)}{D(x)} \Rightarrow \int \frac{N(x)}{D(x)}\, dx = \int \left[Q(x) + \frac{R(x)}{D(x)}\right] dx = \int Q(x)\, dx + \int \frac{R(x)}{D(x)}\, dx$$

2. $$N(x) = f'(D(x))$$

$$\int \frac{f'(x)}{f(x)}\, dx = ln|f(x)| + c$$

3. $$deg\big(D(x)\big) = 1$$

$$a \neq 0$$

$$\int \frac{1}{ax+b}\, dx = \frac{1}{a} \int \frac{a}{ax+b}\, dx = \frac{1}{a} ln|ax+b| + c$$

4. $$deg\big(D(x)\big) = 2$$

$$a \neq 0,\ \Delta > 0$$

$$\int \frac{px+q}{ax^2+bx+c}\,dx \Rightarrow (1)\ ax^2+bx+c = a(x-x_1)(x-x_2) \Rightarrow$$

$$(2)\ \frac{px+q}{ax^2+bx+c} = \frac{A}{a(x-x_1)} + \frac{B}{(x-x_2)} \Rightarrow$$

$$(3)\ \int \frac{px+q}{ax^2+bx+c}\,dx = \int \left[\frac{A}{a(x-x_1)} + \frac{B}{(x-x_2)}\right] dx$$

$$a \neq 0,\ \Delta = 0$$

$$\int \frac{px+q}{ax^2+bx+c}\,dx \Rightarrow (1)\ ax^2+bx+c = a(x-x_1)^2 \Rightarrow$$

$$(2)\ \frac{px+q}{ax^2+bx+c} = \frac{A}{a(x-x_1)} + \frac{B}{(x-x_1)^2} \Rightarrow$$

$$(3)\ \int \frac{px+q}{ax^2+bx+c}\,dx = \int \left[\frac{A}{a(x-x_1)} + \frac{B}{(x-x_1)^2}\right] dx$$

$$Note:\ if\, p = 0$$

$$\int \frac{q}{ax^2+bx+c}\,dx = \int \frac{q}{a(x-x_1)^2}\,dx = -\frac{q}{a(x-x_1)} + c$$

5.

$$a \neq 0,\ \Delta < 0$$

$$CASE\,1:\ \int \frac{1}{ax^2+bx+c}\,dx \Rightarrow$$

$$(1)\ gather\, x^2{}'s\, coefficient:\ \frac{1}{a}\int \frac{1}{x^2+mx+n}\,dx$$

$$(2)\ D(x) = (x+h)^2 + k^2$$

$$(3)\ \int \frac{1}{ax^2+bx+c}\,dx = \frac{1}{a}\int \frac{1}{(x+h)^2+k^2}\,dx = \frac{1}{ak}\,arctg\frac{x+h}{k} + c$$

$$a \neq 0, p \neq 0,\ \Delta < 0$$

$$CASE\,2: \int \frac{px+q}{ax^2+bx+c}\,dx \Rightarrow$$

$$(1) \int \frac{px+q}{ax^2+bx+c}\,dx = r\int \frac{2ax+b}{ax^2+bx+c} + s\int \frac{1}{ax^2+bx+c}$$

$$(2) \int \frac{2ax+b}{ax^2+bx+c} = \int \frac{f'(x)}{f(x)} = ln|ax^2+bx+c| + c$$

$$(3) \int \frac{1}{ax^2+bx+c}\,dx = \frac{1}{a}\int \frac{1}{(x+h)^2+k^2}\,dx = \frac{1}{ak}\,arctg\frac{x+h}{k} + c$$

$$\frac{1}{(x+h)^2+k^2}\,dx = \frac{1}{ak}\,arctg\frac{x+h}{k} + c$$

$$(4)\ Then\,add\,the\,results$$

5 Sum

5.1 Property

1. $$\sum_{n=0}^{\infty}(an+bn) = \sum_{n=0}^{\infty} an + \sum_{n=0}^{\infty} bn$$

2. $$\sum_{n=0}^{\infty} k\, an = k \sum_{n=0}^{\infty} an$$

3. if the sum is definitely non-negative terms

$$\sum_{n=0}^{\infty} an = \begin{cases} convergent \\ divergent\, positively \end{cases}$$

4. When you have a sum:

$$\sum_{n=0}^{\infty} an$$

You have to study the sum $\Leftrightarrow lim_{x\to+\infty} an = 0$

5.2 Notable Sum

1. Harmonic Sum

$$\sum_{n=1}^{\infty} \frac{1}{n} = \text{convergent}$$

2. Generalized Harmonic Sum

$$\sum_{n=1}^{\infty} \frac{1}{n^{\alpha}} = \begin{cases} \alpha > 1 \Rightarrow convergent \\ 0 < \alpha \leq 1 \Rightarrow divergent \end{cases}$$

3. Mengoli Sum (Telescopic)

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)}\ or\ \sum_{n=0}^{\infty} \frac{1}{n} - \frac{1}{n+1} = convergent$$

4. Harmonic Sum with alternating signs

$$\sum_{n=1}^{\infty} \frac{1}{n}(-1)^{(n-1)} = convergent$$

5. Telescopic Bis

$$\sum_{n=1}^{\infty} 1 + \frac{1}{n} = divergent^{+}$$

6. Geometric Sum

$$\sum_{n=0}^{\infty} q^n = \begin{cases} |q| < 1 \\ \Rightarrow\ convergentq \geq 1 \Rightarrow divergent \\ q \leq -1 \Rightarrow irregolar \end{cases}$$

7.
$$\sum_{n=1}^{\infty} \frac{1}{n^{\alpha}(ln\, n)^{\beta}} = \begin{cases} \alpha > 1\, or\, \alpha = 1\, and\, \beta > 1 \Rightarrow convergent \\ \alpha < 1\, or\, \alpha = 1\, and\, \beta \leq 1 \Rightarrow divergent \end{cases}$$

5.3 Non-negative Terms Sum

1. Ratio test (D'Alembert's criterion)

$$if\, \exists lim_{x\to\infty} \frac{an+1}{an} = l \Rightarrow \sum_{n=1}^{\infty} an \to \begin{cases} l < 1 \Rightarrow convergent \\ l > 1 \Rightarrow divergent \\ l = 1 \Rightarrow changecriterion \end{cases}$$

2. Root test (Cauchy's criterion)

$$if\, \exists lim_{x\to\infty} \sqrt[n]{an} = l \Rightarrow \sum_{n=1}^{\infty} an \to \begin{cases} l < 1 \Rightarrow convergent \\ l > 1 \Rightarrow divergent \\ l = 1 \Rightarrow changecriterion \end{cases}$$

3. I Comparison criterion

   Supposed 0$\leq an \leq bn$
   $if\ \sum bn\, converges\ \Rightarrow \sum an\, converges$
   $if\ \sum an\, diverges^{+} \Rightarrow \sum bn\, diverges^{+}$

4. II Comparison criterion

   Supposed an,bn$\geq 0\, bn \neq 0$
   $$if\, \exists lim_{x\to\infty} \tfrac{an}{bn} = l \Rightarrow \begin{cases} if\, bn\, converges\, and\, 0 < l < \infty \Rightarrow an\, converges \\ if\, bn\, diverges\, and\, l > 0\, (or\ \pm\infty) \Rightarrow an\, diverges \end{cases}$$

5. Criterion of asymptotic comparison

   Supposed an¿0 Definitively and bn¿0 Definitively

$$if\, \exists lim_{x\to\infty} \frac{an}{bn} = 1 \Rightarrow \sum an\, and \sum bn \text{ they have the same behavior}$$

6. Integral criterion

   Supposed f:[1,+∞[$\to \mathbb{R}\, a\, positive\, and\, increasing\, function$
   $\sum_{n=1}^{\infty} an\, converges \Leftrightarrow \int_1^{\infty} f(x)\, dx\, converges$

   $\sum_{n=1}^{\infty} an\, diverges \Leftrightarrow \int_1^{\infty} f(x)\, dx\, diverges$

7. Raabe's criterion

   Supposed
$$\sum_{n=1}^{\infty} an$$
   a positive term sum
$$if\, \exists lim_{x\to\infty}\, n\left(\frac{an}{an+1} - 1\right) = l \Rightarrow \sum_{n=1}^{\infty} an \to \begin{cases} l < 1 \Rightarrow convergent \\ l > 1 \Rightarrow divergent \\ l = 1 \Rightarrow indetermined \end{cases}$$

5.4 Variable Terms Sum

1. Absolute convergence

$$if\, converges \sum_{n=0}^{\infty} |an| \rightarrow \sum an\, converges$$

2. Leibniz's criterion
   Suppose an is a succession

$$if \left\{ \begin{array}{l} 0\, an \geq 0\, Definitively \\ an \rightarrow 0\, for\, n \rightarrow \infty \\ an + 1 \leq an\, Definitively \end{array} \right\} \Rightarrow \sum_{n=0}^{\infty} (-1)^n an\, converges$$